

Deformation Theory of Embedded Curves and Its Application

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The slides for all four lectures are available via this QR code.

Plan of Talk

- September 17 (Day 1)
 - ① Basics of deformation theory
 - ② Mumford's example
- September 18 (Day 2)
 - ③ Obstructedness criterion and applications
 - ④ Recent results on deformations of curves in $S_{2,2,2} \subset \mathbb{P}^5$

Deformation Theory of Embedded Curves and Its Applications

Lecture 1: Basics of Deformation Theory

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Plan of the lecture series

- **Day 1**

- ① Basics of deformation theory

- ② Mumford's example

- **Day 2**

- ③ Obstructedness criterion and applications

- ④ Recent results on deformations of curves in $S_{2,2,2} \subset \mathbb{P}^5$

Guiding question for Lecture 1

What does the Hilbert scheme look like near a point representing an embedded curve?

Starting point: moving a curve

Let

$$C \subset \mathbb{P}^n$$

be a smooth embedded curve.

The most basic geometric question is:

How can C move inside the ambient projective space $\mathbb{P}^n$?

We want to distinguish:

- actual families of curves;
- infinitesimal motions;
- tangent directions which do not come from actual motions.

The Hilbert scheme as an organizing space

The Hilbert scheme turns the problem of moving curves into the local geometry of a parameter space.

$$C \subset \mathbb{P}^n \quad \Longleftrightarrow \quad [C] \in \mathbf{Hilb}_P(\mathbb{P}^n).$$

A family of embedded curves corresponds to a path, or more generally a morphism, into the Hilbert scheme.

Thus the question becomes:

What is the local structure of $\mathbf{Hilb}_P(\mathbb{P}^n)$ near $[C]$?

Hilbert polynomial

For a projective scheme $X \subset \mathbb{P}^n$ over $k = \bar{k}$, set

$$\mathcal{O}_X(m) := \mathcal{O}_{\mathbb{P}^n}(m) \otimes \mathcal{O}_X.$$

The *Hilbert polynomial* is

$$P_X(t) := \chi(X, \mathcal{O}_X(t)).$$

For a curve $C \subset \mathbb{P}^n$ of degree d and genus g ,

$$P_C(t) = dt + 1 - g.$$

Role of P

The Hilbert polynomial fixes the numerical type of the embedded scheme.

Grothendieck's Hilbert scheme

Theorem 1 (Grothendieck'61[1])

Fix a polynomial P . There exists a projective scheme

$$\mathbf{Hilb}_P(\mathbb{P}^n)$$

parametrizing closed subschemes $X \subset \mathbb{P}^n$ with $P_X = P$.

It carries a universal family

$$\mathcal{W} \subset \mathbb{P}^n \times \mathbf{Hilb}_P(\mathbb{P}^n),$$

flat over $\mathbf{Hilb}_P(\mathbb{P}^n)$, whose fiber over $[X]$ is X .

Why flatness appears

Flatness is the condition that makes the fibers vary continuously.

- In a flat projective family, the Hilbert polynomial is constant.
- Therefore flat families with Hilbert polynomial P are exactly the families controlled by $\mathbf{Hilb}_P(\mathbb{P}^n)$.
- Infinitesimal deformations are infinitesimal flat families.

Deformation theory studies the local geometry of this parameter space.

Space curves: a motivating example

For smooth connected curves $C \subset \mathbb{P}^3$ of degree d and genus g , write

$$\mathcal{H}_{d,g} := \{[C] \in \mathbf{Hilb}_{dt+1-g}(\mathbb{P}^3) \mid C \text{ smooth and connected}\}.$$

Classical results show that these schemes can have many different behaviors:

empty or non-empty	Castelnuovo-type bounds
irreducible in some ranges	Ein's theorem
highly reducible	many components may occur
non-reduced	Mumford's example

Point of the example

Even for smooth space curves, the Hilbert scheme can have complicated local geometry. Deformation theory provides tools to detect it.

Local geometry near a point

Let

$$H := \mathbf{Hilb}_P(\mathbb{P}^n), \quad [X] \in H.$$

The local behavior at $[X]$ may be:

smooth	all infinitesimal motions integrate locally
singular	tangent space larger than the local dimension
non-reduced	nilpotent directions are present

Our goal is to measure this local behavior by cohomology of the normal bundle.

The tangent space: a microscope

Let $(\mathcal{O}_{H,[X]}, \mathfrak{m})$ be the local ring of H at $[X]$. The Zariski tangent space is

$$T_{[X]}H = (\mathfrak{m}/\mathfrak{m}^2)^\vee.$$

Equivalently,

$$T_{[X]}H \simeq \{\mathrm{Spec} k[\varepsilon]/(\varepsilon^2) \rightarrow H \text{ through } [X]\}.$$

Interpretation

Tangent vectors are first-order ways of moving X in $\mathbb{P}^n$.

First-order embedded deformations

Put

$$D_1 := \operatorname{Spec} k[\varepsilon]/(\varepsilon^2).$$

A first-order embedded deformation of $X \subset \mathbb{P}^n$ is a closed subscheme

$$X_1 \subset \mathbb{P}^n \times D_1$$

such that

- $X_1 \rightarrow D_1$ is flat;
- $X_1 \times_{D_1} \operatorname{Spec} k \simeq X$.

By the universal property of the Hilbert scheme, this is the same as a tangent vector at $[X]$.

Main fact 1: tangent directions

Let $\mathcal{I}_X \subset \mathcal{O}_{\mathbb{P}^n}$ be the ideal sheaf of X , and define the normal sheaf by

$$N_{X/\mathbb{P}^n} := \mathcal{H}om(\mathcal{I}_X/\mathcal{I}_X^2, \mathcal{O}_X).$$

Theorem 2 (cf. [2, 4])

There is a natural isomorphism

$$T_{[X]} \mathbf{Hilb}_P(\mathbb{P}^n) \simeq \mathbf{Hom}_{\mathcal{O}_{\mathbb{P}^n}}(\mathcal{I}_X, \mathcal{O}_X) \simeq H^0(N_{X/\mathbb{P}^n}).$$

NB: No local complete intersection assumption is needed here. If X is l.c.i., then $N_{X/\mathbb{P}^n}$ is locally free, so it is the usual normal bundle.

Geometric meaning of the normal sheaf

For any closed subscheme $X \subset \mathbb{P}^n$,

$$T_{[X]}H \simeq H^0(N_{X/\mathbb{P}^n}).$$

A section of $N_{X/\mathbb{P}^n}$ records how the defining equations of X change to first order.

When X is l.c.i., this is literally a global normal vector field along X :

first-order embedded deformations
= sections of the normal sheaf

Example: a line in $\mathbb{P}^3$

Let $L \simeq \mathbb{P}^1$ be a line in $\mathbb{P}^3$. Its normal bundle is

$$N_{L/\mathbb{P}^3} \simeq \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(1).$$

Hence

$$h^0(N_{L/\mathbb{P}^3}) = 2 + 2 = 4, \quad H^1(N_{L/\mathbb{P}^3}) = 0.$$

Therefore

$$\dim T_{[L]} \mathbf{Hilb}_{t+1}(\mathbb{P}^3) = 4.$$

Geometric check

Lines in $\mathbb{P}^3$ are parametrized by the Grassmannian $G(1, 3)$, which is smooth of dimension 4.

Example: a hypersurface

Let $X = (f = 0) \subset \mathbb{P}^n$ be a smooth hypersurface of degree d . Then

$$N_{X/\mathbb{P}^n} \simeq \mathcal{O}_X(d).$$

Let g be another homogeneous polynomial of degree d . A first-order deformation is written as

$$f + \varepsilon g = 0.$$

Changing g by a multiple of f gives the same section of $\mathcal{O}_X(d)$, so the deformation is recorded by

$$\bar{g} \in H^0(X, \mathcal{O}_X(d)) = H^0(N_{X/\mathbb{P}^n}).$$

Bridge to the local calculation

This is the simplest model for the rule $f \mapsto \bar{g}$ in $\mathbf{Hom}(I, A/I)$.

Local picture behind the theorem

Work locally with

$$V = \operatorname{Spec} A \subset \mathbb{P}^n, \quad X \cap V = \operatorname{Spec} A/I.$$

Let us see where the group $\operatorname{Hom}_A(I, A/I)$ comes from.

A first-order deformation is given by an ideal

$$I_1 \subset A[\varepsilon]/(\varepsilon^2)$$

reducing to I modulo ε .

For each $f \in I$, flatness allows a lift

$$f_1 = f + \varepsilon g \in I_1.$$

Taking g modulo I gives

$$\varphi(f) := \bar{g} \in A/I.$$

Why φ is well defined

Suppose that the same $f \in I$ has two lifts in I_1 ,

$$f + \varepsilon g, \quad f + \varepsilon g'.$$

Their difference is $\varepsilon(g - g') \in I_1$. Flatness gives

$$g - g' \in I,$$

so $\bar{g} = \bar{g}'$ in A/I .

Moreover, for $a \in A$,

$$a(f + \varepsilon g) = af + \varepsilon ag \in I_1,$$

and therefore

$$\varphi(af) = \bar{a} \varphi(f).$$

Thus the deformation determines

$$\varphi \in \mathbf{Hom}_A(I, A/I).$$

Conversely: from φ to a deformation

Given

$$\varphi \in \mathbf{Hom}_A(I, A/I),$$

define

$$I_\varphi = \{ f + \varepsilon g \mid f \in I, \bar{g} = \varphi(f) \} \subset A[\varepsilon]/(\varepsilon^2).$$

Then I_φ reduces to I modulo ε , and

$$A[\varepsilon]/(\varepsilon^2)/I_\varphi$$

is flat over $k[\varepsilon]/(\varepsilon^2)$.

Hence locally

$$\{\text{first-order embedded deformations}\} \longleftrightarrow \mathbf{Hom}_A(I, A/I).$$

These local descriptions glue to

$$T_{[X]} \mathbf{Hilb}_P(\mathbb{P}^n) \simeq H^0(N_{X/\mathbb{P}^n}).$$

The next question

A tangent vector gives a deformation over

$$D_1 = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2).$$

But an actual family should be visible to higher order.

Can we lift it to

$$D_2 := \operatorname{Spec} k[\varepsilon]/(\varepsilon^3)?$$

In other words, given

$$X_1 \subset \mathbb{P}^n \times D_1,$$

does there exist a flat

$$X_2 \subset \mathbb{P}^n \times D_2$$

with $X_2 \times_{D_2} D_1 = X_1$?

From local liftings to an obstruction

Suppose $X \subset \mathbb{P}^n$ is l.c.i. A first-order deformation can be lifted locally to second order:

$$X_1|_{U_i} \rightsquigarrow X_2^{(i)}.$$

The problem is global: on an overlap U_{ij} , two local second-order liftings may differ.

These differences form a Čech 1-cocycle with values in the normal sheaf:

$$\{\text{failure on } U_{ij}\} \in Z^1(N_{X/\mathbb{P}^n}).$$

Idea

The obstruction is the cohomology class of this gluing failure.

Obstructions

Let

$$\varphi \in H^0(N_{X/\mathbb{P}^n})$$

be the tangent vector corresponding to X_1 .

Theorem 3 (cf. [4, 2, 3])

Assume from now on that $X \subset \mathbb{P}^n$ is l.c.i. Then there is a primary obstruction class

$$\mathbf{ob}(\varphi) (= \varphi \cup e \cup \varphi) \in H^1(N_{X/\mathbb{P}^n})$$

such that

$$X_1 \text{ lifts to second order} \iff \mathbf{ob}(\varphi) = 0.$$

First-order motion $\xrightarrow{\mathbf{ob}}$ obstruction space.

Why $H^1(N)$ appears

Local liftings are always possible if $X \subset \mathbb{P}^n$ is a l.c.i.

The real issue is whether the local liftings agree on overlaps.

- first-order deformations are global sections of N ;
- incompatibilities on overlaps form a Čech 1-cocycle;
- the obstruction class is its class in $H^1(N)$.

Geometric meaning

$H^1(N_{X/\mathbb{P}^n})$ measures the failure of local higher-order motions to glue globally.

Čech calculation: first-order compatibility

Let $X \subset \mathbb{P}^n$ be l.c.i. of codimension 2 and choose an affine open cover $\{U_i\}$ of $\mathbb{P}^n$ such that

$$I_i = (f_{i1}, f_{i2}), \quad \mathbf{f}_i = \begin{pmatrix} f_{i1} \\ f_{i2} \end{pmatrix}.$$

On U_{ij} choose a matrix M_{ij} with

$$\mathbf{f}_j = M_{ij} \mathbf{f}_i.$$

Here the matrices $\overline{M_{ij}} \pmod{I_X}$ are invertible and give the transition matrices of $N_{X/\mathbb{P}^n}^\vee = \mathcal{I}_X / \mathcal{I}_X^2$. Write the first-order deformation locally as

$$\mathbf{f}_i + \varepsilon \mathbf{g}_i.$$

Compatibility on overlaps means that, for some matrix N_{ij} ,

$$\mathbf{f}_j + \varepsilon \mathbf{g}_j = (M_{ij} + \varepsilon N_{ij})(\mathbf{f}_i + \varepsilon \mathbf{g}_i) \pmod{\varepsilon^2}.$$

Comparing coefficients of ε gives

$$\mathbf{g}_j = M_{ij} \mathbf{g}_i + N_{ij} \mathbf{f}_i.$$

Čech calculation: trying to lift to second order

We try to lift the local equations to

$$\mathbf{f}_i + \varepsilon \mathbf{g}_i + \varepsilon^2 \mathbf{h}_i \pmod{\varepsilon^3}.$$

On U_{ij} this requires matrices P_{ij} such that

$$\mathbf{f}_j + \varepsilon \mathbf{g}_j + \varepsilon^2 \mathbf{h}_j = (M_{ij} + \varepsilon N_{ij} + \varepsilon^2 P_{ij}) \cdot (\mathbf{f}_i + \varepsilon \mathbf{g}_i + \varepsilon^2 \mathbf{h}_i) \pmod{\varepsilon^3}.$$

Comparing coefficients of ε^2 and restricting to X gives

$$\overline{\mathbf{h}_j} - \overline{M_{ij}} \overline{\mathbf{h}_i} = \overline{N_{ij} \mathbf{g}_i}. \quad (2)$$

Thus the question is whether the right-hand side of (2) is a Čech coboundary.

The obstruction cocycle

Define a **1**-cochain

$$\beta_{ij} \in \Gamma(U_{ij} \cap X, N_{X/\mathbb{P}^n})$$

by

$$\beta_{ij}(\mathbf{f}_i) = \overline{N_{ij}\mathbf{g}_i}.$$

The collection $\beta = \{\beta_{ij}\}$ is a Čech **1**-cocycle.

Meaning of the terms

- $\mathbf{g}_i$ is the first-order data,
- $\mathbf{h}_i$ is the second-order correction,
- $N_{ij}\mathbf{g}_i$ is the failure to glue.

The obstruction class

The cocycle $\beta = \{\beta_{ij}\}$ determines

$$[\beta] \in H^1(X, N_{X/\mathbb{P}^n}).$$

Equation (2) has a solution $\{\mathbf{h}_i\}$ exactly when

$$[\beta] = 0.$$

Therefore

$$X_1 \text{ lifts over } k[\varepsilon]/(\varepsilon^3) \iff [\beta] = 0.$$

This class is the obstruction to gluing the local second-order liftings.

Identifying the cocycle with $\varphi \cup e \cup \varphi$

Let

$$0 \rightarrow \mathcal{I}_X \rightarrow \mathcal{O}_{\mathbb{P}^n} \rightarrow \mathcal{O}_X \rightarrow 0$$

have extension class $e \in \text{Ext}^1(\mathcal{O}_X, \mathcal{I}_X)$. The associated coboundary sends

$$\varphi \in \text{Hom}(\mathcal{I}_X, \mathcal{O}_X) \quad \text{to} \quad \delta(\varphi) = \varphi \cup e.$$

The local lifts of φ differ on overlaps by $N_{ij}\mathbf{f}_i$. Thus equation (1) gives

$$\delta(\varphi)_{ij}(\mathbf{f}_i) = N_{ij}\mathbf{f}_i.$$

Hence

$$\varphi_i(N_{ij}\mathbf{f}_i) = \overline{N_{ij}\mathbf{g}_i} = \beta_{ij}(\mathbf{f}_i).$$

Thus

$$[\beta] = \text{ob}(\varphi) = \varphi \cup e \cup \varphi.$$

If X is l.c.i., this class lies in $H^1(N_{X/\mathbb{P}^n})$.

What vanishing tells us

The obstruction theory gives a useful sufficient condition.

$$H^1(N_{X/\mathbb{P}^n}) = 0 \implies \mathbf{Hilb}_P(\mathbb{P}^n) \text{ is smooth at } [X].$$

But the converse is false as a logical statement:

$$H^1(N_{X/\mathbb{P}^n}) \neq 0 \not\Rightarrow [X] \text{ is obstructed.}$$

Obstruction space $\neq$ actual obstruction

The obstruction space may be nonzero even when every actual obstruction class vanishes. To prove obstructedness, one must find a tangent direction whose obstruction class is actually nonzero.

Type-checking the cup product

The formula is a Yoneda product:

$$\mathcal{I}_X \xrightarrow{\varphi} \mathcal{O}_X \xrightarrow{e} \mathcal{I}_X[1] \xrightarrow{\varphi} \mathcal{O}_X[1].$$

Thus

$$\varphi \cup e \cup \varphi \in \mathrm{Ext}^1(\mathcal{I}_X, \mathcal{O}_X).$$

Equivalently, the coboundary map associated with

$$0 \rightarrow \mathcal{I}_X \rightarrow \mathcal{O}_{\mathbb{P}^n} \rightarrow \mathcal{O}_X \rightarrow 0$$

gives

$$\delta(\varphi) = \varphi \cup e \in \mathrm{Ext}^1(\mathcal{I}_X, \mathcal{I}_X), \quad \mathrm{ob}(\varphi) = \delta(\varphi) \cup \varphi.$$

Summary of Lecture 1

- 1 The Hilbert scheme organizes embedded deformations of $X \subset \mathbb{P}^n$.
- 2 Its tangent space at $[X]$ is the space of first-order embedded deformations:

$$T_{[X]} \mathbf{Hilb}_P(\mathbb{P}^n) \simeq H^0(N_{X/\mathbb{P}^n}).$$

- 3 A tangent direction may fail to extend to second order.
- 4 For l.c.i. X , this failure is measured by

$$\mathrm{ob}(\varphi) = \varphi \cup e \cup \varphi \in H^1(N_{X/\mathbb{P}^n}).$$

Question for Lecture 2

Can $\text{ob}(\varphi) \neq 0$ actually occur for a smooth space curve $C \subset \mathbb{P}^3$?

Mumford's answer

Yes. Lecture 2 explains how a specific smooth space curve produces nonzero obstructions and hence a generically non-reduced component of the Hilbert scheme.

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Deformation Theory of Embedded Curves and Its Applications

Lecture 2: Mumford's Example

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Plan of the lecture series

- **Day 1**

- ① Basics of deformation theory
- ② Mumford's example

- **Day 2**

- ③ Obstructedness criterion and applications
- ④ Recent results on deformations of curves in $S_{2,2,2} \subset \mathbb{P}^5$

Question from Lecture 1

Can $\text{ob}(\varphi) \neq 0$ actually occur for a smooth space curve?

From Lecture 1 to Lecture 2

Lecture 1 gave two basic facts for an embedded curve $C \subset \mathbb{P}^3$:

$$T_{[C]} \mathbf{Hilb}(\mathbb{P}^3) \simeq H^0(C, N_{C/\mathbb{P}^3}), \quad \text{ob}(\alpha) \in H^1(C, N_{C/\mathbb{P}^3}).$$

The important point was:

$$H^1(N_{C/\mathbb{P}^3}) \neq 0 \quad \not\Rightarrow \quad C \text{ is obstructed.}$$

Today we look at Mumford's example, where an actual tangent direction has a nonzero primary obstruction.

The Hilbert scheme of space curves

We write

$$H_{d,g}^S := \{[C] \in \mathbf{Hilb}_{dt+1-g}(\mathbb{P}^3) \mid C \text{ smooth and connected}\}.$$

This is the open subscheme parametrizing smooth connected space curves of degree d and genus g .

In this lecture we focus on

$$H_{14,24}^S.$$

Mumford's theorem

Theorem 1 (Mumford'62 [1])

The Hilbert scheme $H_{14,24}^S$ contains an irreducible component W of dimension **56** such that

$H_{14,24}^S$ is non-reduced at the generic point of W .

Equivalently, $H_{14,24}^S$ is **generically non-reduced** along W .

Goal

Construct W , explain the numerical mystery **56** versus **57**, and exhibit a nonzero obstruction.

The key numerical mystery

For a general curve $C \in W$, we will have

$$\dim W = 56$$

but

$$\dim T_{[C]} H_{14,24}^S = h^0(N_{C/\mathbb{P}^3}) = 57.$$

Thus there is one extra tangent direction.

Guiding questions

Where does this extra tangent direction come from?

Why does it not integrate to an actual extra family of curves?

The curves in Mumford's component

Let $S \subset \mathbb{P}^3$ be a smooth cubic surface, and let $E \subset S$ be a line.

Put

$$h := \mathcal{O}_S(1), \quad C \sim 4h + 2E.$$

A general curve $C \in |4h + 2E|$ is smooth and connected.

The basic configuration is

$$C \subset S \subset \mathbb{P}^3, \quad S \text{ cubic}, \quad E \subset S \text{ a line.}$$

Intersection numbers on a cubic surface

Recall that a smooth cubic surface is isomorphic to the blow-up of $\mathbb{P}^2$ at six points:

$$S \simeq \mathbf{Bl}_{6 \text{ pts}} \mathbb{P}^2.$$

With $h = \mathcal{O}_S(1)$ and E a line, we have

$$h^2 = 3, \quad h.E = 1, \quad E^2 = -1, \quad K_S = -h.$$

These four numbers determine the degree and genus of $C \sim 4h + 2E$.

Degree and genus of C

Since $C \sim 4h + 2E$, we get

$$d = C.h = (4h + 2E).h = 12 + 2 = 14.$$

Also

$$C^2 = (4h + 2E)^2 = 16h^2 + 16h.E + 4E^2 = 48 + 16 - 4 = 60.$$

By adjunction,

$$g(C) = \frac{(C + K_S).C}{2} + 1 = \frac{60 - 14}{2} + 1 = 24.$$

Thus $[C] \in H_{14,24}^S$.

The family W

Let $\mathcal{U}$ be the open subset of $|\mathcal{O}_{\mathbb{P}^3}(3)|$ parametrizing smooth cubic surfaces.

For each $S \in \mathcal{U}$ and each line $E \subset S$, take a general curve

$$C \in |4h + 2E|.$$

The curves obtained in this way form a locally closed irreducible family

$$W \subset H_{14,24}^S.$$

For a general $C \in W$, the cubic surface S containing C is unique.

Dimension of W : the number 56

The space of cubic surfaces in $\mathbb{P}^3$ has dimension

$$\dim |O_{\mathbb{P}^3}(3)| = \binom{3+3}{3} - 1 = 19.$$

On S , Riemann–Roch gives

$$\chi(O_S(C)) = \frac{C(C - K_S)}{2} + 1 = \frac{60 + 14}{2} + 1 = 38.$$

Since $h^i(O_S(C)) = 0$ ($i > 0$) and $|C|$ is nonempty,

$$\dim |C| = 37.$$

Therefore

$$\dim W = 19 + 37 = 56.$$

For a general $C \in W$, the containing cubic surface is unique, since two distinct cubics meet in degree 9.

Computing the tangent space

Consider the normal bundle sequence

$$0 \rightarrow N_{C/S} \rightarrow N_{C/\mathbb{P}^3} \rightarrow N_{S/\mathbb{P}^3}|_C \rightarrow 0.$$

Since C is a divisor on S and S is cubic,

$$N_{C/S} \simeq \mathcal{O}_C(C), \quad N_{S/\mathbb{P}^3}|_C \simeq \mathcal{O}_C(3).$$

Thus

$$0 \rightarrow \mathcal{O}_C(C) \rightarrow N_{C/\mathbb{P}^3} \rightarrow \mathcal{O}_C(3) \rightarrow 0,$$

where $H^1(C, \mathcal{O}_C(C)) \simeq H^1(C, -K_S|_C + K_C) = 0$ by adjunction and ampleness of $-K_S \sim \mathcal{O}_S(1)$.

The obstruction space is one-dimensional

From the normal bundle sequence one obtains

$$H^1(N_{C/\mathbb{P}^3}) \simeq H^1(O_C(3)).$$

By Serre duality and adjunction,

$$H^1(O_C(3)) \simeq H^0(K_C - 3h)^\vee.$$

Since

$$K_C = (K_S + C)|_C = (3h + 2E)|_C,$$

we get

$$H^0(K_C - 3h)^\vee \simeq H^0(O_C(2E))^\vee \simeq H^0(O_S(2E))^\vee \simeq k.$$

Hence

$$h^1(N_{C/\mathbb{P}^3}) = h^1(O_C(3)) = 1.$$

The tangent space has dimension 57

For a smooth space curve of degree d , one has

$$\chi(N_{C/\mathbb{P}^3}) = 4d.$$

In our case $d = 14$, so

$$\chi(N_{C/\mathbb{P}^3}) = 4d = 56.$$

Since $h^1(N_{C/\mathbb{P}^3}) = 1$,

$$h^0(N_{C/\mathbb{P}^3}) = \chi(N_{C/\mathbb{P}^3}) + h^1(N_{C/\mathbb{P}^3}) = 56 + 1 = 57.$$

Therefore

$$\dim T_{[C]} H_{14,24}^S = 57.$$

Dimension count and the dichotomy

For a general $C \in W$, we have

$$\dim W = 56, \quad \dim T_{[C]}H_{14,24}^S = 57.$$

Therefore the tangent space is one dimension larger than the constructed family.

Dichotomy

Either W is contained in a larger reduced component of dimension 57, or the Hilbert scheme is non-reduced along W .

To prove Mumford's theorem, it is enough to find

$$\alpha \in H^0(N_{C/\mathbb{P}^3}) \quad \text{with} \quad \text{ob}(\alpha) \neq 0.$$

Where the extra tangent direction comes from

The extra tangent direction is related to the line

$$E \subset S.$$

We will construct a section

$$u \in H^0(C, \mathcal{O}_C(3))$$

with a hidden pole along E , and then lift it to

$$\alpha \in H^0(N_{C/\mathbb{P}^3}).$$

The pole along E produces the nonzero obstruction

Unprojection is a way to understand this rational deformation geometrically.

A cubic surface containing a line

Choose coordinates so that

$$E = (x = y = 0) \subset \mathbb{P}^3.$$

A cubic surface S containing E can be written as

$$S = (Ax + By = 0),$$

where A and B are quadratic forms.

Put

$$f := Ax + By.$$

On S , the relation $Ax + By = 0$ gives the rational section

$$\theta := \frac{A}{y} = -\frac{B}{x}$$

of $\mathcal{O}_S(1)$.

The rational section θ

The projection $\pi_E : S \setminus E \rightarrow \mathbb{P}^1$ from E is given by $(x : y)$, and its extension to E is given by $(-B : A)$.

Lemma 2

π extends to a morphism

$$\Phi_{|h-E|} : S \rightarrow \mathbb{P}^1,$$

and S is covered by $D(x')$ and $D(y')$, where $D(x) = D(x') \setminus E$ and $D(y) = D(y') \setminus E$.

Thus $\theta = A/y = -B/x$ is a global section of

$$\mathcal{O}_S(1)(E) = \mathcal{O}_S(1) \otimes_S \mathcal{O}_S(E).$$

This gives rise to the infinitesimal deformation with a pole that will later produce the obstruction.

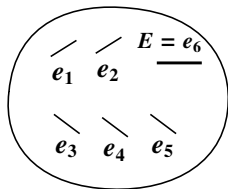
Unprojection of the line (I)

A smooth cubic surface S_3 is obtained by blowing up the quartic del Pezzo surface

$$S_4 = (2) \cap (2) \subset \mathbb{P}^4$$

at one point.

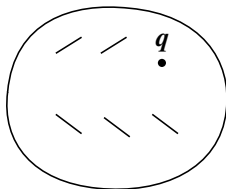
$$S_3 \simeq \text{Bl}_{6 \text{ pts}} \mathbb{P}^2 \subset \mathbb{P}^3$$



blow-down

projection from q

$$S_4 \simeq \text{Bl}_{5 \text{ pts}} \mathbb{P}^2 \subset \mathbb{P}^4$$



Geometric picture

The blow-down contracts the (-1) -curve $E \subset S_3$ to a point $q \in S_4$. Conversely, S_3 is obtained from S_4 by projection from q .

Unprojection of the line (II)

Lemma 3 (Reid'00 [3])

The blow-down of E from S_3 to S_4 is given by

$$\Phi_{|O_S(1)(E)|} : S_3 \longrightarrow \mathbb{P}^4, \quad \mathbf{x} \longmapsto (\mathbf{x}, \theta(\mathbf{x})),$$

which contracts E to a point $q \in S_4$. Moreover, this morphism is the inverse of the projection $\pi_q : S_4 \subset \mathbb{P}^4 \longrightarrow S_3$ from q .

Proof. Introduce a new coordinate t corresponding to θ . Since

$$x\theta = -B, \quad y\theta = A,$$

the image of S_3 in $\mathbb{P}^4$ is defined by

$$xt + B = 0, \quad yt - A = 0. \quad \square$$

Interpretation

θ realizes the unprojection $S_3 \rightarrow S_4 = (2) \cap (2) \subset \mathbb{P}^4$ of E .

The intersection $Z = C \cap E$

Let

$$Z := C \cap E.$$

Since

$$C.E = (4h + 2E).E = 4 - 2 = 2,$$

Z is a length 2 subscheme of the line $E \simeq \mathbb{P}^1$.

Choose a quadratic polynomial

$$\xi \in H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))$$

such that

$$\mathrm{div}(\xi|_E) = Z.$$

The zero of ξ along Z will cancel the pole of θ on C .

A regular section on C

Because θ has a pole along E and ξ vanishes along $Z = C \cap E$, the product restricts regularly to C :

$$u := (\xi\theta)|_C \in H^0(C, \mathcal{O}_C(3)).$$

More precisely,

$$\theta|_C \in H^0(C, \mathcal{O}_C(1)(Z)), \quad \xi|_C \in H^0(C, \mathcal{O}_C(2)(-Z)),$$

and hence

$$(\xi\theta)|_C \in H^0(C, \mathcal{O}_C(3)).$$

This section will be the exterior component of our tangent direction.

Constructing the extra tangent direction

Since

$$N_{S/\mathbb{P}^3}|_C \simeq \mathcal{O}_C(3),$$

the section

$$u = (\xi\theta)|_C \in H^0(C, \mathcal{O}_C(3))$$

is a normal direction to the surface S along C . The normal bundle sequence gives

$$\begin{array}{ccccc} H^0(N_{C/\mathbb{P}^3}) & \rightarrow & H^0(\mathcal{O}_C(3)) & \rightarrow & 0. \\ \psi & & \psi & & \\ \exists \alpha & \mapsto & u & & \end{array}$$

Then u lifts to

$$\alpha \in H^0(N_{C/\mathbb{P}^3}).$$

We will show that this α is obstructed.

Primary obstruction as a cup product

Recall the standard ideal sequence

$$0 \rightarrow \mathcal{I}_C \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_C \rightarrow 0$$

with extension class

$$e \in \mathrm{Ext}^1(\mathcal{O}_C, \mathcal{I}_C).$$

For

$$\alpha \in H^0(N_{C/\mathbb{P}^3}) \simeq \mathrm{Hom}(\mathcal{I}_C, \mathcal{O}_C),$$

the primary obstruction is

$$\mathrm{ob}(\alpha) = \alpha \cup e \cup \alpha \in H^1(N_{C/\mathbb{P}^3}).$$

We now compute a nonzero component of this class.

The cubic equation and the exterior component

The cubic surface S is defined by

$$f = Ax + By.$$

The exterior component of α satisfies

$$\alpha(f) = u = (\xi\theta)|_C.$$

Thus the obstruction is computed by the composition

$$f \xrightarrow{\alpha} u \xrightarrow{\delta} \delta(u) \xrightarrow{\alpha} \text{ob}(\alpha)(f),$$

where

$$\delta : H^0(\mathcal{O}_C(3)) \rightarrow H^1(\mathcal{I}_C(3))$$

is the coboundary map induced by $0 \rightarrow \mathcal{I}_C \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_C \rightarrow 0$.

Computing $\delta(u)$

Work on the cover

$$D(x) \cup D(y).$$

Since

$$\theta = -\frac{B}{x} = \frac{A}{y},$$

the section $u = \xi\theta$ is represented locally by

$$-\frac{\xi B}{x} \quad \text{on } D(x), \quad \frac{\xi A}{y} \quad \text{on } D(y).$$

On $D(x) \cap D(y)$, the relevant component of the Čech coboundary is

$$\delta(u) = \frac{\xi A}{y} - \left(-\frac{\xi B}{x}\right) = \frac{\xi}{xy}(Ax + By) = \frac{\xi}{xy}f.$$

Applying α again

Since $\alpha(f) = u$, the last application of α gives

$$\text{ob}(\alpha)(f) = \alpha(\delta(u)) = \frac{\xi}{xy} u.$$

Thus the obstruction is represented by the cup product

$$t|_C \cup u, \text{ where } t = \left(\frac{\xi}{xy} \right) \text{ in } H^1(\mathcal{O}_S(2E))$$

in $H^1(\mathcal{O}_C(3)(2E))$.

Meaning

The first factor ξ cancels the pole of θ on C , but the second-order calculation still remembers the pole along E .

Nonvanishing of the obstruction

The choice of ξ gives

$$\operatorname{div}(\xi|_E) = Z = C \cap E.$$

Lemma 4 (cf. [2])

On smooth cubic surface $S = (Ax + By = 0)$, the following are equivalent:

- ① $t = 0$ in $H^1(S, \mathcal{O}_S(2E))$, where $t = (\xi/xy)$.
- ② $\xi|_E \in \langle A|_E, B|_E \rangle$,
- ③ $\operatorname{div}(\xi|_E)$ is cut out by the subpencil $\Lambda|_E \simeq \mathbb{P}^1$, where $\Lambda = |h - E|$.

Since C is general in $|4h + 2E|$, Z is general in $|O_E(2)|$, and hence $t \neq 0$ in $H^1(S, \mathcal{O}_S(2E))$. Using the Serre duality pairing, one further proves $t|_C \cup u \neq 0$ in $H^1(C, \mathcal{O}_C(3)(2E))$. Therefore

$\operatorname{ob}(\alpha)(f) \neq 0.$

Conclusion of the obstruction computation

We have constructed

$$\alpha \in H^0(N_{C/\mathbb{P}^3})$$

such that

$$\mathbf{ob}(\alpha)(f) \neq 0.$$

Hence

$$\mathbf{ob}(\alpha) \neq 0 \quad \text{in } H^1(N_{C/\mathbb{P}^3}).$$

Thus C is obstructed as a curve in $\mathbb{P}^3$.

The extra tangent direction does not integrate.

From obstruction to non-reducedness

Recall the numerical facts:

$$\dim W = 56, \quad h^0(N_{C/\mathbb{P}^3}) = 57.$$

The nonzero obstruction shows that the extra tangent direction is not realized by an actual deformation.

Therefore the reduced local dimension remains **56**, while the tangent space has dimension **57**.

Hence $H_{14,24}^S$ is generically non-reduced along W .

What makes the example special?

Mumford's example combines three ingredients:

- 1 a curve $C \sim 4h + 2E$ on a smooth cubic surface S ;
- 2 a line $E \subset S$ producing a rational section

$$\theta \in H^0(S, \mathcal{O}_S(h + E));$$

- 3 a first-order deformation whose obstruction detects the pole of θ along E .

The line E is the geometric source of the pathology.

Summary of Lecture 2

- 1 Mumford's curves are general members of

$$|4h + 2E|$$

on a smooth cubic surface $S \subset \mathbb{P}^3$ containing a line E .

- 2 They have

$$d = 14, \quad g = 24.$$

- 3 They form a family W of dimension 56.
- 4 At a general point of W ,

$$h^0(N_{C/\mathbb{P}^3}) = 57.$$

- 5 The extra tangent direction has nonzero primary obstruction.

Take-home message

Mumford's example is the fundamental example where a smooth space curve represents a generically non-reduced point of the Hilbert scheme.

The mechanism is:

line on a cubic surface $\implies$ rational deformation with a pole
 $\implies$ nonzero primary obstruction $\implies$ non-reduced Hilbert scheme.

Next lecture

Mumford's example suggests a broader question:

How can we systematically detect nonzero obstructions for curves lying on a surface inside a threefold?

In Lecture 3 we will discuss an obstructedness criterion and its applications.

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Deformation Theory of Embedded Curves and Its Applications

Lecture 3: Obstructedness Criterion and Applications

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KU Lectures in Algebraic Geometry 2026

Plan of the lecture series

- **Day 1**

- ① Basics of deformation theory
- ② Mumford's example

- **Day 2**

- ③ **Obstructedness criterion and applications**
- ④ Recent results on deformations of curves in $S_{2,2,2} \subset \mathbb{P}^5$

Guiding question for Lecture 3

What was essential in Mumford's example, and how can we use it systematically?

What was essential in Mumford's example?

Lecture 2 was not just a special calculation. The essential pattern was

$$C \subset S \subset \mathbb{P}^3, \quad E \subset S \text{ a line,}$$

together with a rational deformation of S having a pole along E .

$$E \implies \text{deformation with a pole} \implies \text{nonzero obstruction.}$$

Today's aim

We formulate this pattern as a criterion, then use Hilbert–flag schemes to convert obstructed tangent directions into information on Hilbert schemes.

The basic geometric setup

Let

$$C \subset S \subset X$$

where

- X is a smooth projective threefold,
- $S \subset X$ is a smooth surface,
- $C \subset S$ is a smooth curve,
- $E \subset S$ is a smooth curve with $E^2 < 0$.

We study embedded deformations of

$$C \subset X.$$

Question

Which tangent directions in $H^0(N_{C/X})$ have nonzero obstruction?

Exterior component of a deformation

The inclusion $C \subset S \subset X$ gives

$$0 \rightarrow N_{C/S} \rightarrow N_{C/X} \xrightarrow{\pi_{C/S}} N_{S/X}|_C \rightarrow 0.$$

Hence a first-order deformation

$$\alpha \in H^0(N_{C/X})$$

has an exterior component

$$u := \pi_{C/S}(\alpha) \in H^0(N_{S/X}|_C).$$

Interpretation

u records the part of the deformation that moves C out of the surface S .

Allowing a pole along E

In Mumford's example the exterior component came from a section with a pole. We now impose the same structure abstractly.

$$\beta \in H^0(S, N_{S/X}(E)), \quad \beta|_C = \bar{u} \in H^0(C, N_{S/X}(E)|_C).$$

Here $\bar{u}$ is the image of

$$u \in H^0(N_{S/X}|_C) \quad \text{in} \quad H^0(N_{S/X}(E)|_C).$$

Guiding image

The deformation is regular on C , but it is explained by a rational deformation of S with a controlled pole along E .

The diagram behind the construction

We use the natural restriction maps

$$\begin{array}{ccc} H^0(N_{S/X}) & \longrightarrow & H^0(N_{S/X}(E)) \\ \downarrow & & \downarrow \\ H^0(N_{S/X}|_C) & \longrightarrow & H^0(N_{S/X}(E)|_C). \end{array}$$

The tangent direction α gives

$$u = \pi_{C/S}(\alpha) \in H^0(N_{S/X}|_C).$$

The criterion starts from the situation where

$$\bar{u} = \beta|_C \quad \text{for some} \quad \beta \in H^0(N_{S/X}(E)).$$

The divisor Δ

Define

$$\Delta := C + K_X|_S - 2E.$$

This divisor measures the residual part of the configuration after the pole contribution along E is removed. In Mumford's example,

$$X = \mathbb{P}^3, \quad K_X|_S = -4h, \quad C \sim 4h + 2E,$$

and hence

$$\Delta = C + K_X|_S - 2E = 0.$$

Role of Δ

The conditions on Δ ensure that the pole along E is detected by the obstruction class.

Before the theorem: three roles

The criterion has three types of hypotheses.

- ① **Numerical balance:** the divisor Δ has exactly the right degree on E .
- ② **Local nontriviality:** the pole deformation is nontrivial along E .
- ③ **Global survival:** sections on E lift from S , so the local nonvanishing cannot disappear globally.

Relation with Mumford

The second condition is the abstract replacement for the nonvanishing class $[\xi/(xy)] \neq 0$ in Lecture 2.

Obstructedness criterion

Theorem 1 (Mukai–Nasu'09 [1])

Let $\alpha \in H^0(N_{C/X})$, and suppose that

$$\overline{\pi_{C/S}(\alpha)} = \beta|_C \quad \text{in } H^0(N_{S/X}(E)|_C)$$

for some $\beta \in H^0(N_{S/X}(E))$. Then $\text{ob}(\alpha) \neq 0$ if the following three conditions hold:

- 1 $\Delta \cdot E = 2(-E^2 + g(E) - 1)$;
- 2 $\beta|_E \cup \mathbf{k}_E \neq 0$ in $H^1(E, \mathcal{O}_E(2E))$;
- 3 $H^0(S, \Delta) \rightarrow H^0(E, \Delta|_E)$ is surjective.

The extension class on E

The class $\mathbf{k}_E$ comes from the normal bundle sequence of

$$E \subset S \subset X :$$

$$0 \rightarrow N_{E/S} \rightarrow N_{E/X} \rightarrow N_{S/X}|_E \rightarrow 0.$$

Pairing the section

$$\beta|_E \in H^0(N_{S/X}(E)|_E)$$

with the extension class gives

$$\beta|_E \cup \mathbf{k}_E \in H^1(E, N_{E/S}(E)) = H^1(E, \mathcal{O}_E(2E)).$$

Meaning

This is the local obstruction created by the pole along E .

Mumford revisited: condition (1)

In Mumford's example,

$$X = \mathbb{P}^3, \quad S = (Ax + By = 0), \quad E = (x = y = 0), \quad C \sim 4h + 2E.$$

We have

$$\Delta = C + K_X|_S - 2E = 0.$$

Since $E \simeq \mathbb{P}^1$ and $E^2 = -1$,

$$\Delta \cdot E = 0 = 2(-E^2 + g(E) - 1).$$

Thus the numerical condition holds.

Mumford revisited: condition (2)

The normal bundle sequence of $E \subset S \subset \mathbb{P}^3$ is

$$0 \rightarrow \mathcal{O}_E(-1) \rightarrow \mathcal{O}_E(1)^{\oplus 2} \rightarrow \mathcal{O}_E(3) \rightarrow 0.$$

The exterior map is represented by

$$(A|_E, B|_E).$$

For the tangent direction constructed in Lecture 2,

$$\beta|_E = \xi|_E \in H^0(\mathcal{O}_E(2)).$$

The local nonvanishing condition becomes

$$\beta|_E \cup \mathbf{k}_E = \left[\frac{\xi}{xy} \right]_E \neq 0 \quad \text{in } H^1(\mathcal{O}_E(2E)).$$

Mumford revisited: condition (3)

Since $\Delta = \mathbf{0}$, the restriction map in condition (3) is simply

$$H^0(S, \mathcal{O}_S) \longrightarrow H^0(E, \mathcal{O}_E),$$

which is surjective. Therefore all three conditions of the criterion are satisfied.

Conclusion

The criterion recovers the nonzero obstruction in Mumford's example without repeating the full Cech calculation.

Mumford revisited: from computation to criterion

Lecture 2 gave the explicit calculation

$$\delta(u) = \frac{\xi}{xy}f, \quad \mathbf{ob}(\alpha)(f) = \frac{\xi}{xy}u.$$

Lecture 3 replaces this by three geometric checks:

numerical balance + local nontriviality + global survival.

From example to method

Mumford's example is not an isolated trick; it is the first instance of a general obstruction mechanism.

From one obstructed curve to a component

The criterion gives a tangent direction α with

$$\mathbf{ob}(\alpha) \neq 0.$$

To prove that an entire component is non-reduced, we combine it with a dimension count:

$$\dim W < h^0(N_{C/X}).$$

Strategy

Construct a family W of curves on surfaces, show that the only extra tangent directions are exterior, and prove that those exterior directions are obstructed.

Application to uniruled threefolds

Applying the criterion to many uniruled threefolds gives the following type of result.

Theorem 2 (Mukai–Nasu'09 [1])

Let X be a projective threefold over an algebraically closed field. Suppose that

- ① there is a line $E \simeq \mathbb{P}^1 \subset X$ such that $N_{E/X}$ is globally generated;
- ② there is a smooth surface $S \subset X$ such that
 - ① E is a (-1) -curve on S ;
 - ② $p_g(S) = h^1(N_{S/X}) = 0$.

Then $\mathbf{Hilb}^{sc} X$ contains a generically non-reduced component.

Idea of the construction

Blow down the (-1) -curve

$$\varepsilon : S \rightarrow \bar{S}.$$

Take a sufficiently ample divisor Δ on $\bar{S}$, and consider a general curve

$$C \in |\varepsilon^* \Delta - K_X|_S + 2E|.$$

Such curves form a family W in $\mathbf{Hilb}^{sc} X$. The construction is arranged so that

$$h^0(N_{C/X}) = \dim W + 1.$$

The criterion shows that the extra tangent directions are obstructed.

The dimension-count strategy

The strategy is the same as in Mumford's example.

$$\dim W < h^0(N_{C/X}).$$

This gives extra tangent directions. The obstructedness criterion shows that, for exterior tangent directions

$$\alpha \in H^0(N_{C/X}) \setminus T_{W,C},$$

one has

$$\text{ob}(\alpha) \neq 0.$$

Conclusion

The reduced local dimension is $\dim W$, but the tangent space is larger. Thus the Hilbert scheme is generically non-reduced along W .

Fano threefolds of index two

As a consequence, one obtains generically non-reduced components for many smooth Fano threefolds of index two.

Examples

$$X_3 = (3) \subset \mathbb{P}^4, \quad X_4 = (2) \cap (2) \subset \mathbb{P}^5,$$

$$X_5 = \mathrm{Gr}(2, 5) \cap \mathbb{P}^6, \quad \dots$$

These are del Pezzo threefolds. The existence of suitable pairs $E \subset S \subset X$ makes the criterion applicable.

A later refinement: K3 surfaces in Fano threefolds

There is also a more recent form of the criterion for curves on a K3 surface inside a Fano threefold.

$$C \subset S_{K3} \subset X_{Fano3}.$$

The goal is again to prove that

$$C \subset X$$

is obstructed by finding a negative curve $E \subset S$ and a tangent direction whose exterior component has a controlled pole along E .

Motivation

In this setting the surface containing the curve varies, so Hilbert–flag schemes become especially useful.

A K3/Fano criterion

Theorem 3 (Nasu'17 [2])

Let $C \subset S \subset X$ with S a K3 surface in a Fano threefold X . Suppose that

- ① $H^1(N_{(C,S)/X}) = 0$;
- ② $D := C + K_X|_S \geq 0$, $D^2 \geq 0$, and $D \cdot E = -2$ for some $E \simeq \mathbb{P}^1$ on S ;
- ③ $H^1(S, D - 3E) = 0$;
- ④ the map $\pi_{E/S}(E)$ is not surjective.

Then C is obstructed in X .

In particular, $\mathbf{Hilb}^{sc} X$ is singular at $[C]$.

Why Hilbert–flag schemes?

The criterion detects obstructed exterior directions, but to understand the local dimension of the Hilbert scheme we also need to control the directions coming from surfaces containing C .

$$C \subset S \subset X.$$

The Hilbert scheme remembers only C . The Hilbert–flag scheme remembers the pair (C, S) .

Guiding idea

Study the easier local geometry of pairs (C, S) , then forget S to obtain information about the Hilbert scheme of curves.

Hilbert–flag scheme

Definition 4

Let X be a projective scheme, and let P, Q be Hilbert polynomials. The Hilbert–flag scheme

$$\mathbf{HF}_{P,Q}^X$$

parametrizes pairs of closed subschemes

$$C \subset S \subset X$$

such that

$$P_C = P, \quad P_S = Q.$$

It is a projective scheme.

Natural projections

There are natural morphisms

$$\begin{array}{ccc} & \mathbf{HF}_{P,Q}^X & \\ \swarrow \scriptstyle pr_1 & & \searrow \scriptstyle pr_2 \\ \mathbf{Hilb}_P X & & \mathbf{Hilb}_Q X \end{array}$$

The point

$$(C, S) \in \mathbf{HF}_{P,Q}^X$$

maps to

$$[C] \in \mathbf{Hilb}_P X, \quad [S] \in \mathbf{Hilb}_Q X.$$

Idea

The map p_1 forgets the surface; the map p_2 forgets the curve.

The normal sheaf of the pair (C, S)

The tangent space to the Hilbert–flag scheme is described by a sheaf $N_{(C,S)/X}$, encoding compatible deformations of C and S . For $C \subset S \subset X$, define $N_{(C,S)/X}$ by the Cartesian diagram

$$\begin{array}{ccc} N_{(C,S)/X} & \xrightarrow{\pi_2} & N_{S/X} \\ \downarrow \pi_1 & & \downarrow \text{lc} \\ N_{C/X} & \xrightarrow{\pi_{C/S}} & N_{S/X}|_C. \end{array}$$

Thus

$$N_{(C,S)/X} = N_{C/X} \times_{N_{S/X}|_C} N_{S/X}.$$

Then

$$T_{(C,S)} \mathbf{HF}_{P,Q}^X \simeq H^0(N_{(C,S)/X}).$$

Tangent and obstruction spaces for flags

The sheaf $N_{(C,S)/X}$ plays the same role for the pair (C, S) as $N_{C/X}$ plays for the curve C .

$H^0(N_{(C,S)/X})$ is the tangent space at (C, S) ,

and

$H^1(N_{(C,S)/X})$ is the obstruction space for the flag.

Useful case

If $H^1(N_{(C,S)/X}) = 0$, then the Hilbert–flag scheme is smooth at (C, S) .

Comparing the flag and the Hilbert scheme

The forgetful map

$$pr_1 : \mathbf{HF}_{P,Q}^X \longrightarrow \mathbf{Hilb}_P X, \quad (C, S) \longmapsto [C]$$

has a tangent map

$$p_1 : H^0(N_{(C,S)/X}) \longrightarrow H^0(N_{C/X}).$$

Its kernel consists of infinitesimal deformations of S which keep C fixed:

$$\ker(p_1) \simeq H^0(N_{S/X}(-C)).$$

Key point

If $H^0(N_{S/X}(-C)) = 0$, then the flag tangent space embeds into the Hilbert tangent space.

A lemma on local dimensions

Lemma 5 (Nasu'23 [3])

We have

$$\dim_{(C,S)} \mathbf{HF}_{P,Q}^X = \dim_{[C]} \mathbf{Hilb}_P X$$

if the following hold:

- ① $H^1(N_{(C,S)/X}) = 0$ and $H^0(N_{S/X}(-C)) = 0$;
- ② for every tangent direction

$$\alpha \notin \operatorname{im} \left[H^0(N_{(C,S)/X}) \rightarrow H^0(N_{C/X}) \right],$$

one has $\mathbf{ob}(\alpha) \neq 0$.

Meaning

The flag scheme gives the actual reduced directions. Every tangent direction outside the flag image is killed by an obstruction.

From one obstruction to all extra directions

The obstructedness criterion first gives a way to prove

$$\mathbf{ob}(\alpha) \neq 0$$

for a chosen exterior tangent direction. To determine a component, we need this for every tangent direction not explained by the flag:

$$H^0(N_{C/X}) = \mathbf{im} H^0(N_{(C,S)/X}) \oplus \{\text{extra exterior directions}\}.$$

Strategy

- 1 Use the Hilbert–flag scheme to compute the local dimension coming from pairs (C, S) .
- 2 Use the criterion to obstruct all extra exterior directions.

Then the Hilbert scheme has no additional reduced directions beyond those coming from the flag.

The general pattern

The pattern is now clear:

$$C \subset S \subset X, \quad E \subset S \text{ a negative curve.}$$

- 1 Construct a family W of curves on surfaces.
- 2 Use the Hilbert–flag scheme to compute the expected local dimension.
- 3 Identify exterior tangent directions.
- 4 Use the criterion to prove these directions are obstructed.

Outcome

This produces singular or non-reduced behavior of the Hilbert scheme.

Summary of Lecture 3

- 1 Mumford's calculation reflects a general mechanism: a pole along a negative curve can create a nonzero obstruction.
- 2 The obstructedness criterion detects this using

$$\beta|_E \cup k_E \neq 0.$$

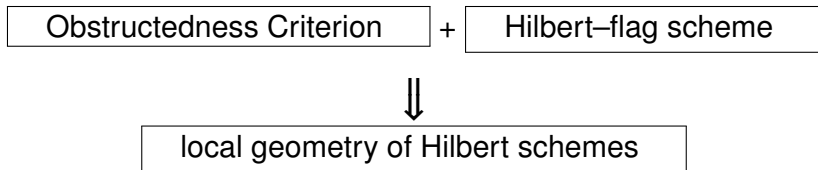
- 3 The criterion applies to many curves on surfaces inside threefolds.
- 4 Hilbert–flag schemes help control the local dimension of the Hilbert scheme.

Key message

Obstructions can be detected geometrically, not only by direct Cech calculation.

Next lecture

Lecture 3 gives the method:



In Lecture 4, we apply this philosophy to curves lying on

$$S_{2,2,2} \subset \mathbb{P}^5,$$

and discuss recent results on deformations of curves on complete intersection K3 surfaces.

Next guiding question

What new phenomena appear for curves on $S_{2,2,2} \subset \mathbb{P}^5$?

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Deformation Theory of Embedded Curves and Its Applications

Lecture 4: Recent Results on Deformations of Curves in $S_{2,\dots,2} \subset \mathbb{P}^{n+2}$

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Tokai University

September 18, 2026

KU Lectures in Algebraic Geometry 2026

Plan of the lecture series

- **Day 1**

- ① Basics of deformation theory
- ② Mumford's example

- **Day 2**

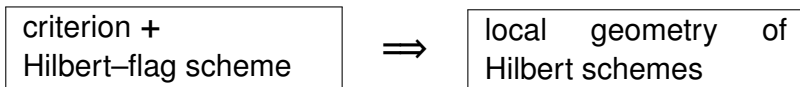
- ③ Obstructedness criterion and applications
- ④ Recent results on deformations of curves in $S_{2,2,2} \subset \mathbb{P}^5$

Guiding question for Lecture 4

What new phenomena appear for curves on complete intersection K3 surfaces?

From Lecture 3 to Lecture 4

Lecture 3 gave the method:



We now apply this philosophy to curves on complete intersections of quadrics.

Today's focus

Curves on

$$S_{2,\dots,2} \subset \mathbb{P}^{n+2},$$

especially the K3 case

$$S_{2,2,2} \subset \mathbb{P}^5.$$

Motivation: Hilbert schemes of curves

Let X be a projective variety over an algebraically closed field and write

$$\mathbf{Hilb}^{\mathrm{sc}} X := \{[C] \in \mathbf{Hilb} X \mid C \subset X \text{ smooth and connected}\}.$$

Problem

Construct generically non-reduced components of $\mathbf{Hilb}^{\mathrm{sc}} X$ for a given variety X .

The difficulty is to exhibit tangent directions which are not realized by actual deformations.

Known sources of non-reduced components

The problem has a long history.

X	examples and constructions
$\mathbb{P}^3$	Mumford, Gruson–Peskine, Kleppe, Ellia, ...
$\mathbb{P}^n$ ($n \geq 4$)	Ciliberto–Lopez–Miranda, Choi–Iliev–Kim, N., ...
Fano threefolds	Mukai–N., N.

New direction

What can be said for higher-dimensional algebraic varieties?

A result for Fano threefolds

Lecture 3 discussed the following type of result.

Theorem 1 (Mumford'62 [2], Mukai-N.'09 [1], N.'19 [3])

Let X be a smooth Fano threefold of Picard number one. Then $\mathbf{Hilb}^{\mathrm{sc}} X$ contains a **generically non-reduced** component, whose general member C is contained in a **del Pezzo surface** or **K3 surface**

$$S \subset X$$

containing **a line or a conic** E .

What was known in dimension three

In dimension three, the construction uses surfaces containing a negative curve E .

index	surface S	curve class
≥ 2	del Pezzo surface	$-K_X _S + 2E$
1	K3 surface	$-2K_X _S + 2E$

Question

Can one find a similar phenomenon for higher-dimensional ambient varieties?

Another motivation: local dimension

There is also a more local question.

Problem

Determine the local dimension of $\mathbf{Hilb}^{\text{sc}} X$ at a point $[C]$.

This becomes especially subtle when

$$g(C) > 0.$$

In that case, a naive parameter count is often not enough, and one must control all extra tangent directions.

Tool

The Hilbert–flag scheme helps compare deformations of C with deformations of pairs (C, S) .

Complete intersections of quadrics

Let

$$S = S_{2,\dots,2} \subset \mathbb{P}^{n+2} = \mathbb{P}$$

be a smooth complete intersection of n quadrics. We study smooth connected curves

$$C \subset S \subset \mathbb{P}.$$

Goal

Understand deformations of C in the ambient projective space $\mathbb{P}$.

Small values of n

The behavior depends strongly on n .

n	$S_{2,\dots,2}$	behavior
1	$S_2 \simeq \mathbb{P}^1 \times \mathbb{P}^1$	curves are unobstructed
2	$S_{2,2} \simeq \mathbf{Bl}_{p_1,\dots,p_5} \mathbb{P}^2$	obstructions controlled by $e_1, \dots, e_5$ (N.'26 [5])
≥ 3	K3 or general type	new phenomena

Today

We focus on $n \geq 3$, especially $n = 3$.

The basic setting

Assume

$$n \geq 3, \quad m \geq 1,$$

and let

$$S = S_{2,\dots,2} \subset \mathbb{P}^{n+2}$$

be smooth. Suppose that S contains a line

$$E \subset S.$$

Put

$$h := \mathcal{O}_S(1), \quad D := (n-1)h + E \in \mathrm{Pic} S.$$

We consider a general curve

$$C \in |mD|.$$

Why the divisor $D = (n - 1)h + E$?

The linear system

$$|D| = |O_S(n - 1)(E)|$$

is related to the contraction of the line E . Since

$$E^2 = 1 - n, \quad D.E = (n - 1)h.E + E^2 = 0,$$

the morphism defined by $|D|$ contracts E .

Guiding picture

The divisor D is the pull-back of an ample divisor on the contracted surface.

Contraction and vanishing

Let

$$f : S \longrightarrow \bar{S}$$

be the contraction morphism defined by $|D|$. Then there is an ample divisor Δ on $\bar{S}$ such that

$$D = f^* \Delta.$$

Therefore, for $m \gg 0$,

$$H^1(S, \mathcal{O}_S(mD)) = 0$$

by Serre vanishing on $\bar{S}$.

Why this matters

The curves $C \in |mD|$ form a well-controlled family, even though $D.E = 0$.

The main theorem: common structure

Let $p = \text{char } k \geq 0$.

Theorem 2 (N.'26[4])

Assume that m is sufficiently large and, if $p > 0$, that $p \nmid m$. Let C be a general member of $|mD|$. Then:

- 1 C is smooth and connected;
- 2 the Hilbert–flag scheme $\mathbf{HF} \mathbb{P}$ is nonsingular at (C, S) of dimension

$$d_{n,m} := \frac{n(n+2)(n+3)}{2} + 2n + 2 + \dim |mD|.$$

Point

The theorem first gives a clean family of pairs $C \subset S \subset \mathbb{P}^{n+2}$ with controlled flag deformations.

Where the dimension $d_{n,m}$ comes from

The number

$$d_{n,m} = \frac{n(n+2)(n+3)}{2} + 2n + 2 + \dim |mD|$$

has a simple meaning.

data	contribution
choice of the line E	$2n + 2$
choice of $S_{2,\dots,2}$ containing E	$\frac{n(n+2)(n+3)}{2}$
choice of $C \in mD $	$\dim mD $

Interpretation

This is the expected local dimension coming from pairs $C \subset S \subset \mathbb{P}^{n+2}$.

The main theorem: the dichotomy

The same construction gives two different outcomes.

K3 case: $n = 3$

The curves C , with E and S varying, form a **generically non-reduced** component

$$W \subset \mathbf{Hilb}^{\mathrm{sc}} \mathbb{P}^5.$$

Moreover,

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Higher-dimensional case: $n > 3$

The corresponding family gives a **generically smooth** component of dimension $d_{n,m}$.

The headline phenomenon

The theorem says that the complete intersections of quadrics split into two regimes:

$$n = 3:$$

generically non-reduced

whereas

$$n > 3:$$

generically smooth

Thus the K3 surface

$$S_{2,2,2} \subset \mathbb{P}^5$$

is exactly the exceptional case in this construction.

Question

What mechanism produces the extra non-reduced directions only in the K3 case?

The K3 case made explicit

For $n = 3$ we have

$$S = S_{2,2,2} \subset \mathbb{P}^5, \quad D = 2h + E.$$

For a general curve

$$C \in |m(2h + E)|,$$

one obtains

$$\deg C = 17m, \quad g(C) = 17m^2 + 1.$$

The family W has dimension

$$\dim W = d_{3,m} = g(C) + 53.$$

Three extra tangent directions

In the K3 case,

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Thus there are three tangent directions beyond the directions coming from the family W .

Meaning of the number 3

These three directions are the genuine pole directions along the line E .

The goal is to show that all of them are obstructed.

Why exactly three?

The difference comes from infinitesimal deformations with a pole along E . Since

$$N_{S/\mathbb{P}^5} \simeq \mathcal{O}_S(2)^{\oplus 3},$$

and, for a line $E \subset S_{2,2,2}$,

$$N_{S/\mathbb{P}^5}(E)|_E \simeq \mathcal{O}_E^{\oplus 3},$$

the quotient

$$\frac{H^0(N_{S/\mathbb{P}^5}(E))}{H^0(N_{S/\mathbb{P}^5})}$$

has dimension 3.

Question before Theorem 3

Are all genuine pole directions obstructed?

Infinitesimal deformation with a pole

Now assume

$$n = 3, \quad S = S_{2,2,2} \subset \mathbb{P}^5.$$

Let

$$S^\circ := S \setminus E, \quad \mathbb{P}^\circ := \mathbb{P}^5 \setminus E.$$

The open immersion $\iota : S^\circ \hookrightarrow S$ gives inclusions

$$N_{S/\mathbb{P}^5}(mE) \hookrightarrow \iota_* N_{S^\circ/\mathbb{P}^\circ}.$$

Thus a section with a pole along E gives a first-order deformation of the open surface $S^\circ \subset \mathbb{P}^\circ$.

The pole-direction quotient

Twisting by E , we have the exact sequence

$$0 \rightarrow N_{S/\mathbb{P}^5} \rightarrow N_{S/\mathbb{P}^5}(E) \rightarrow N_{S/\mathbb{P}^5}(E)|_E \rightarrow 0.$$

Since

$$N_{S/\mathbb{P}^5}(E)|_E \simeq \mathcal{O}_E^{\oplus 3},$$

the cokernel of

$$H^0(N_{S/\mathbb{P}^5}) \rightarrow H^0(N_{S/\mathbb{P}^5}(E))$$

is three-dimensional.

Interpretation

The quotient records first-order deformations with a genuine simple pole along E .

Theorem 3: every pole direction is obstructed

Theorem 3 (N.'26[4])

Let $S = S_{2,2,2} \subset \mathbb{P}^5$ be a smooth complete intersection containing a line E . For every genuine pole direction

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E)) \setminus H^0(N_{S/\mathbb{P}^5}),$$

the induced first-order deformation β° of $S^\circ \subset \mathbb{P}^\circ$ has nonzero obstruction:

$$\text{ob}(\beta^\circ) \neq 0 \quad \text{in } H^1(N_{S^\circ/\mathbb{P}^\circ}).$$

Geometric meaning of Theorem 3

A section

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E))$$

represents a deformation of S which is allowed to have a simple pole along E .

Statement in words

If the pole is genuine, then the first-order deformation of the open surface $S \setminus E$ cannot be lifted to second order.

Role in Theorem 2

Theorem 3 obstructs all three extra tangent directions in the K3 case.

Equations of S containing a line

Choose coordinates so that

$$E = (X_0 = X_1 = X_2 = X_3 = 0) \subset \mathbb{P}^5.$$

A complete intersection of three quadrics containing E can be written as

$$S = (q_1 = q_2 = q_3 = 0),$$

where

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = A \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

for a 3×4 matrix A of linear forms.

The rational section θ

Let A_i be the 3×3 matrix obtained by deleting the i -th column of A , and put

$$\Delta_i = (-1)^i \det A_i.$$

Then on S we have the rational identity

$$\theta := \frac{\Delta_0}{X_0} = \frac{\Delta_1}{X_1} = \frac{\Delta_2}{X_2} = \frac{\Delta_3}{X_3}.$$

Key point

θ is a rational section of $\mathcal{O}_S(2)$, however, it is a global section of $\mathcal{O}_S(2)(E)$.

Computing the obstruction

The explicit description of θ allows one to compute the primary obstruction

$$\mathbf{ob}(\beta) = \beta \cup e \cup \beta.$$

More precisely, one obtains

$$\mathbf{ob}(\beta) \in H^1(N_{S/\mathbb{P}^5}(3E))$$

and shows that this class is nonzero whenever

$$\beta \notin H^0(N_{S/\mathbb{P}^5}).$$

Message

The pole along E is remembered by the second-order obstruction.

Constructing the family for $n = 3$

Take a line

$$E \subset \mathbb{P}^5, \quad [E] \in \mathbf{Gr}(2, 6).$$

Choose a general complete intersection

$$S = S_{2,2,2} \subset \mathbb{P}^5$$

containing E . Since

$$h^0(\mathbb{P}^5, \mathcal{I}_E(2)) = \binom{7}{2} - h^0(E, \mathcal{O}_E(2)) = 21 - 3 = 18,$$

the pairs (E, S) form an open subset of a Grassmannian bundle over $\mathbf{Gr}(2, 6)$.

Dimension of the pair (E, S)

The line E varies in

$$\mathbf{Gr}(2, 6), \quad \mathbf{dim\,Gr}(2, 6) = 8.$$

For fixed E , the three quadrics defining S span a 3-dimensional subspace of

$$H^0(\mathbb{P}^5, \mathcal{I}_E(2)), \quad \mathbf{dim} = 18.$$

Hence the fiber has dimension

$$\mathbf{dim\,Gr}(3, 18) = 3(18 - 3) = 45.$$

Thus

$$\mathbf{dim}\{(E, S)\} = 8 + 45 = 53.$$

The curves $C \in |m(2h + E)|$

Now take

$$C \in |m(2h + E)|$$

general on S . Then

$$\deg C = 17m, \quad g(C) = 17m^2 + 1.$$

The curves obtained by varying C , E , and S form a family

$$W \subset \mathbf{Hilb}^{\mathrm{sc}} \mathbb{P}^5.$$

Since $\dim\{(E, S)\} = 53$, we have

$$\dim W = g(C) + 53 = d_{3,m}.$$

Cohomology behind the tangent excess

The key input is the exact sequence

$$0 \rightarrow N_{S/\mathbb{P}^5}(-C) \rightarrow N_{S/\mathbb{P}^5}(E - C) \rightarrow N_{S/\mathbb{P}^5}(E - C)|_E \rightarrow 0.$$

Since $N_{S/\mathbb{P}^5} \simeq \mathcal{O}_S(2H)^{\oplus 3}$ and $C \sim m(2H + E)$, we have

$$N_{S/\mathbb{P}^5}(E - C) \simeq \mathcal{O}_S(-(m - 1)D)^{\oplus 3}.$$

For $m \gg 0$, the middle term has no H^0 and no H^1 , while

$$N_{S/\mathbb{P}^5}(E - C)|_E \simeq \mathcal{O}_E^{\oplus 3}.$$

Hence

$$H^1(N_{S/\mathbb{P}^5}(-C)) \simeq k^3.$$

Computing tangent directions

For $m \gg 0$, one obtains exact sequences comparing the flag tangent space and the Hilbert tangent space:

$$H^0(N_{(C,S)/\mathbb{P}^5}) \longrightarrow H^0(N_{C/\mathbb{P}^5}) \longrightarrow k^3 \longrightarrow 0.$$

Therefore

$$h^0(N_{C/\mathbb{P}^5}) = \dim W + 3.$$

Interpretation

The three-dimensional quotient is exactly the space of pole directions along E .

Obstructing the three extra directions

Let

$$\alpha \in H^0(N_{C/\mathbb{P}^5})$$

be a tangent direction outside the image of the flag tangent space.
Its exterior component lifts to a pole deformation

$$\beta \in H^0(N_{S/\mathbb{P}^5}(E)) \setminus H^0(N_{S/\mathbb{P}^5}).$$

The obstruction theorem for $S \setminus E$ gives

$$\mathbf{ob}(\beta^\circ) \neq 0.$$

By restriction, this implies

$$\mathbf{ob}(\alpha) \neq 0.$$

Conclusion in the K3 case

We have

$$\dim W = d_{3,m}, \quad h^0(N_{C/\mathbb{P}^5}) = d_{3,m} + 3.$$

The three extra tangent directions are all obstructed.

Conclusion

W is a generically non-reduced component of $\mathbf{Hilb}^{\text{sc}} \mathbb{P}^5$.

This gives a new family of generically non-reduced components coming from curves on the complete intersection K3 surface $S_{2,2,2} \subset \mathbb{P}^5$.

Summary of Lecture 4

- 1 We studied curves $C \subset S_{2,\dots,2} \subset \mathbb{P}^{n+2}$.
- 2 The divisor $D = (n-1)h + E$ is tied to the contraction of the line E .
- 3 In the K3 case $S_{2,2,2} \subset \mathbb{P}^5$, pole deformations along E produce three extra tangent directions.
- 4 These directions have nonzero obstruction.
- 5 Hence $\mathbf{Hilb}^{\text{sc}} \mathbb{P}^5$ has a generically non-reduced component along the family of curves $C \in |m(2h + E)|$.

Key message

The pole mechanism from Mumford's example persists on complete intersection K3 surfaces.

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